

Terminal 1

```
docker network create fintech-net
```

```
docker compose -f docker-compose.core.yml up -d
```

```
docker ps
```

```
docker logs -f fintech-postgres
```

Terminal 2

```
docker exec -it fintech-postgres psql -U fintech -d fintech_db -c "\dt"
```

```
docker compose -f docker-compose.analytics.yml up -d
```

```
docker exec -it fintech-postgres psql -U fintech -d fintech_db -c "ALTER TABLE fraud_alerts ADD COLUMN IF NOT EXISTS merchant_category TEXT;"
```

Terminal 3

```
python producers/transaction_producer.py
```

Terminal 4

```
docker exec -it fintech-spark-master bash -lc "mkdir -p /tmp/ivy && /opt/spark/bin/spark-submit --master spark://spark-master:7077 --conf spark.jars.ivy=/tmp/ivy --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.5.1,org.postgresql:postgresql:42.7.3 /opt/spark-apps/stream_fraud_detection.py"
```

Terminal 5

```
docker exec -it fintech-postgres psql -U fintech -d fintech_db -c "select count(*) from raw_transactions;"
```

```
docker exec -it fintech-postgres psql -U fintech -d fintech_db -c "select count(*) from fraud_alerts;"
```